

A HYBRID FINITE ELEMENT METHOD FOR NEAR BODIES OF REVOLUTION

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Abstract - A generic formulation for a hybrid finite element solution for three-dimensional electromagnetic scattering is given using the equivalent current approach. The major computational tasks involved in monostatic scattering calculations are analyzed and compared as a function of the method of implementing the near field radiation condition, i.e. Method of Moments (MOM), modal expansion (Mie series), body of revolution (BOR). A new method utilizing a BOR formulation is given which addresses these computational issues. This BOR implementation utilizes Hermite cubic basis functions and a variable number of modes per basis function in order to achieve the greatest efficiency. The combined field integral equation formulation is employed to eliminate non-physical resonances of the mesh boundary. Examples are given showing the efficiency and accuracy of this BOR code by itself, and as part of this new hybrid finite element method.

INTRODUCTION

Finite element methods applied to electromagnetic scattering problems are receiving a great deal of attention due to their ability to model complicated geometry, conducting bodies and penetrable material. However, these methods share with other differential based methods, i.e. finite difference time domain (FDTD), the difficulty in specifying a purely radiating boundary condition at the mesh boundary. Local boundary conditions which maintain sparsity of the system matrix are generalizations of Sommerfeld's condition, and work best when the mesh boundary is far from the scatterer [1]. This requires a significant increase in unknowns to model the field in between. The general class of hybrid finite element methods (HFEM) couple the differential method to another method, i.e. MOM [2, 3, 4] or modal expansion [5, 6, 7], which acts as a global boundary condition implementing the exact radiation condition in the near field. The efficiency of the various implementations depends strongly upon the shape of the scatterer and mesh boundary, and upon the number of solutions required.

In this paper we analyze a generic hybrid finite element formulation for efficiency in monostatic cross section calculations. Based on this analysis, a new hybrid finite element method is given utilizing a special BOR formulation for the mesh boundary which addresses the efficiency issues and eliminates non-physical resonances of the mesh boundary encountered in some implementations [8]. Numerical examples are given showing the efficiency of the BOR code itself and of the hybrid finite element method based on this code.

FORMULATION

Consider the configuration, Fig. 1, of a perfectly conducting scatterer totally enclosed within a surface Γ , with the volume in between denoted by Ω . We consider here only perfectly conducting bodies in order to expedite the development. The magnetic field in Ω is denoted by $\vec{H}$, and the electric and magnetic currents on Γ are respectively labeled as $\vec{J}$ and $\vec{M}$. The scattering problem involves simultaneously solving Maxwell's equations in Ω for the magnetic field $\vec{H}$, coupling this field to the equivalent currents $\vec{J}$ and $\vec{M}$ on Γ , and ensuring that $\vec{J}$, $\vec{M}$ and the incident field specify a radiating solution.

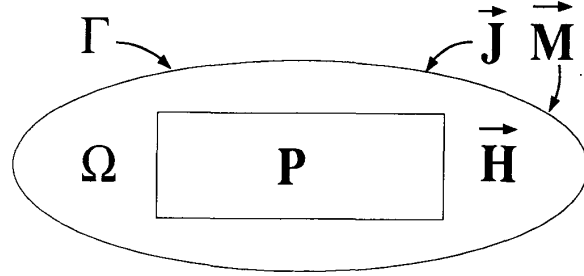


Figure 1: Geometrical configuration for mathematical formulation.

The fields $\vec{H}$, $\vec{J}$, and $\vec{M}$ must satisfy

$$\nabla \times \frac{1}{\epsilon_r} \nabla \times \vec{H}(\vec{r}) - \mu_r k^2 \vec{H}(\vec{r}) = 0 \text{ for } \vec{r} \in \Omega \quad (1)$$

$$\frac{\partial}{\partial \hat{n}} \nabla \times \vec{H}(\vec{r}) = 0 \text{ for } \vec{r} \in P \quad (2)$$

$$\vec{M}(\vec{r}) + \frac{i}{\omega \epsilon_0} \hat{n} \times \nabla \times \vec{H}(\vec{r}) = 0 \text{ for } \vec{r} \in \Gamma \quad (3)$$

$$\vec{J}(\vec{r}) - \hat{n} \times \vec{H}(\vec{r}) = 0 \text{ for } \vec{r} \in \Gamma \quad (4)$$

where ω is the angular frequency, a time dependence $e^{i\omega t}$ is assumed, ϵ_0 and ϵ_r are the free space and relative permittivity respectively, μ_0 and μ_r are the free space and relative permeability respectively, k is the free space propagation constant, and $\hat{n}$ denotes the unit normal to the surfaces.

The boundary condition for a perfect electrical conductor on P (Eq. 2) and for coupling the magnetic current on Γ (Eq. 3) are conditions on the derivative of $\vec{H}$ and are imposed as natural boundary conditions. The boundary condition coupling the electric current on Γ (Eq. 4) is a condition on $\vec{H}$ itself, an essential condition, and must be explicitly enforced. Letting $\vec{T}$ and $\vec{U}$ be a test functions defined on Ω and Γ respectively we write the weak equations

$$\int_{\Omega} [(\nabla \times \vec{T}) \cdot \frac{1}{\epsilon_r} (\nabla \times \vec{H}) - k_0^2 \vec{T} \cdot \mu_r \vec{H}] dV + i\omega \epsilon_0 \int_{\Gamma} \vec{T} \cdot \vec{M} dS = 0 \quad (5)$$

and

$$-i\omega \epsilon_0 \int_{\Gamma} \hat{n} \times \vec{U} \cdot [\vec{J} - \hat{n} \times \vec{H}] dS = 0 \quad (6)$$

where the superscript $*$ is complex conjugation. Integration of Eq. 5 by parts reveals that it is the weak form of Eq. 1 with natural boundary conditions Eq. 2 and Eq. 3, while inspection reveals that Eq. 6 is the weak enforcement of Eq. 4. A finite element discretization of these equations results in matrix equations which we denote by

$$\begin{aligned} [\mathbf{S}] H + [\mathbf{C}] M &= 0 \\ [\mathbf{C}]^* H + [\mathbf{N}] J &= 0 \end{aligned} \quad (7)$$

where the superscript $*$ on matrices refers to the adjoint, i.e. conjugate transpose.

To complete the specification of the scattering problem we introduce the incident field $\vec{H}^i$ and the induced current $\vec{J}^i = 2\hat{n} \times$

$\vec{H}^i$ defined on Γ . The specification of a scattering solution is an equation relating $\vec{J}$, $\vec{M}$ and $\vec{J}^i$ on Γ , such as the magnetic, electric, or combined field integral equations. Discretization will produce a matrix equation which we denote by

$$[\mathbf{K}] J + [\mathbf{L}] M = J^i. \quad (8)$$

Collecting these into a single matrix equation we have the complete system matrix for the problem, Fig. 2.

$\mathbf{S}$		$\mathbf{0}$	$\mathbf{C}$	$\mathbf{H}$	$=$	$\mathbf{0}$
$\mathbf{C}^*$	$\mathbf{N}$	$\mathbf{0}$	$\mathbf{J}$	$\mathbf{0}$		$\mathbf{0}$
$\mathbf{0}$	$\mathbf{K}$	$\mathbf{L}$	$\mathbf{M}$	$\mathbf{J}^i$		$\mathbf{J}^i$

Figure 2: System matrix for generic hybrid finite element formulation.

SOLUTION

The solution of the matrix represented in Fig. 2 leading to efficient monostatic scattering calculations is facilitated by observing that the incident field couples only to $\vec{J}$ and $\vec{M}$, and that these quantities are sufficient for all scattering calculations. Therefore eliminating $\vec{H}$ first, we reduce this system to one involving only $\vec{J}$, $\vec{M}$, and $\vec{J}^i$. Performing this reduction we have

$$[\mathbf{N}] J + [\mathbf{D}] M = 0 \quad (9)$$

$$[\mathbf{K}] J + [\mathbf{L}] M = J^i \quad (10)$$

where

$$[\mathbf{D}] = [-\mathbf{C}^* \mathbf{S}^{-1} \mathbf{C}]. \quad (11)$$

For general three-dimensional scatterers, the matrix $\mathbf{D}$ is *always* a dense matrix, independent of the method of implementing the boundary integral on Γ .

Next, given an incident field J^i ,

$$[\mathbf{L} + \mathbf{K} \mathbf{N}^{-1} \mathbf{D}] M = J^i \quad (12)$$

is solved for M and then

$$[\mathbf{K}] J = J^i - [\mathbf{L}] M \quad (13)$$

is solved for J , after which the scattering may be computed. For monostatic calculations, this is repeated for each monostatic point.

ANALYSIS

The main computational tasks involved in this solution are the factorization and back-substitution of the sparse and dense matrices. To improve efficiency in the sparse calculations, the cost of a single solution must be reduced and the number of sparse solutions required must also be reduced. Independent of solution technique, the sparse matrix task is faster if the sparse matrix is smaller. This requires that the mesh boundary Γ be kept close to the scattering obstacle. The number of sparse solutions required is the number of columns of the matrix $\mathbf{C}$ which equals the total number of degrees of freedom needed to represent $\vec{M}$ or $\vec{J}$.

Reducing this number requires an efficient representation of the electric and magnetic currents on Γ .

Monostatic cross section calculations require many solutions of Eq. 12 and 13, often thousands. The only efficiency gains possible are in reducing the size of the dense matrix, Eq. 11. This again requires an efficient representation of $\vec{J}$ and $\vec{M}$ on Γ .

In summary, efficiency considerations dictate that the mesh boundary should be kept close to the scatterer and that an efficient representation of the electric and magnetic currents be used.

COMPARISON

A modal representation (spherical harmonics) of $\vec{J}$ and $\vec{M}$ requires a spherical mesh boundary Γ . This surface is close to the scatterer only for near spherical scatterers. Due to this restriction the sparse matrix will in general be large because the volume of vacuum between the scatterer and this spherical mesh boundary must be meshed. However, modal expansions are extremely efficient at representing fields on spheres. In addition the scattering integral equation separates completely, and the matrices $\mathbf{K}$ and $\mathbf{L}$ are diagonal making Eq. 13 a triviality. In general, modal methods generate large sparse matrices but are very efficient on the mesh boundary Γ .

A general MOM implementation of the integral equations produces the opposite case. Here, the sparse matrix is minimal or non-existent, since Γ may be general. However, this method requires much higher sampling rates than modal methods to achieve the same accuracy. Also, in this case $\mathbf{K}$ and $\mathbf{L}$ are dense making Eq. 13 a dense system adding to the computational burden. Methods based on MOM generate the smallest sparse matrices but are the least efficient on the mesh boundary.

A BOR formulation is in between these two methods in terms of both sparse matrix size and boundary efficiency. Here, the mesh boundary Γ must be a body of revolution. This is quite flexible and able to conform to a large class of scatterers. Since BOR formulations utilize efficient Fourier modes azimuthally, they also fall in between purely modal and MOM methods in terms of efficiency. The matrices $\mathbf{K}$ and $\mathbf{L}$ are block-diagonal, "in between" dense (MOM) and diagonal (modal). The BOR formulation is a good compromise between sparse matrix size and boundary efficiency, and can be made even better.

HERMITE BOR

Since BOR formulations use highly efficient Fourier modes azimuthally efficiency improvements must come from better representation along the generator. We employ Hermite cubic basis functions and allow the number of modes per basis function to vary over the generator.

Hermite provide continuously differentiable representations, as compared to triangle functions (continuous) or pulse functions (discontinuous). Using these basis functions to also represent the generator of the BOR results in a smooth representation on a smooth surface where the fields themselves are smooth.

Since higher modes are not necessary to represent fields where ρ is small, the number of modes per basis function is determined locally and allowed to vary. This decreases the number of degrees of freedom needed on Γ further reducing the number of sparse solutions required.

We measure efficiency in units of samples per linear (s/λ) or square wavelength (s/λ^2) and convert "modal" terminology to this unit. For example, a circle of radius ρ has circumference $k\rho$ wavelengths, which at 2 samples per wavelength, the Nyquist minimum sampling limit, requires $2k\rho$ samples. In modal expansions, these samples are provided by modes $-k\rho, \dots, 0, \dots, k\rho$,

In terms of this unit, we have found that modal methods require about $3s/\lambda$ ($9s/\lambda^2$), somewhat more than the Nyquist minimum sampling rate. General MOM codes require about $15s/\lambda$ ($225s/\lambda^2$). The number of unknowns, degrees of freedom, per square wavelength is twice the sampling rate due to the two tangential components of the fields on Γ .

Extensive testing of this BOR code revealed that on spheres this method would match the highly efficient Mie series ($9s/\lambda^2$). For more general smooth bodies our sampling guidelines are only $4s/\lambda$ along the generator and $3s/\lambda$ azimuthally. This yields an average minimum sampling rate of only $12s/\lambda^2$.

We also employ the combined field integral equation because Eq. 13 can be singular if the mesh boundary would resonate as a perfect electrical conductor. This results in non-physical resonances in the solution which have occurred in some methods [8].

BOR EXAMPLE

The first example exercises the BOR code itself, computing scattering from a perfectly conducting body of revolution formed by rotating an ellipse of major axis 4λ ($\hat{z}$) and minor axis λ about the z axis. The incident electric field is $\hat{\theta}$ polarized and an "E-plane" bistatic cross section is compared. The sampling rate along the generator averaged only $3.7s/\lambda$. At this sampling rate the Hermite basis functions span more than a wavelength. In comparison, other BOR codes would require at least $10s/\lambda$. Azimuthal sampling was $3s/\lambda$ requiring modes $-5, \dots, +5$. However while the matrices for modes $-3, \dots, +3$ were rank 32, those for ± 4 were rank 28, and ± 5 were rank 20, showing that the highest modes did not couple to the ends of the body where ρ is small. This reduction in 10% in the total number of unknowns is significant in the hybrid finite element method. The sampling rate averaged an extremely low $15.7s/\lambda^2$ over the $10\lambda^2$ of surface area.

Figure 3 shows the results from the Hermite BOR at this low sampling rate, an independent finely sampled calculation from one of the "CICERO" [9] family of codes, both using the combined field equation. Also shown is a result using the magnetic field equation showing the effects of a resonance eliminated by using the combined field equation. This accuracy and efficiency is typical for smooth bodies for which this code is intended.

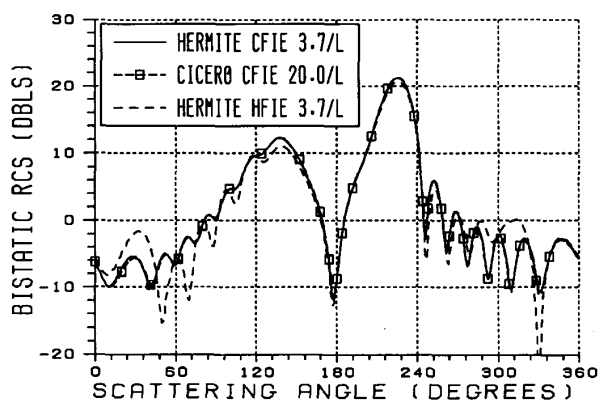


Figure 3: Bistatic scattering (dB/λ^2) from 4λ by λ conducting prolate spheroid illuminated 45° off axis.

HFEM EXAMPLE

This BOR code is the basis for a three-dimensional hybrid finite element method, for which it supplies the exact radiation condition in the near field. The next example exercises this method in computing monostatic scattering from a perfectly conducting box of dimensions $2m$ by $1/2m$ by $1/2m$ at 200 MHz. Figure 4 shows several aspects of this example. The flat plates in the lower left of the figure represent one fourth of the surface of the box. The line running just above these plates is the generator of the BOR. This line stays within about $1/20$ th λ from the box all along its length. This line is then revolved about the $\hat{z}$ axis, generating the BOR mesh boundary which is very close to the scatterer. Figure 4 also shows one fourth of the mesh which occupies the volume between the scatterer and this BOR.

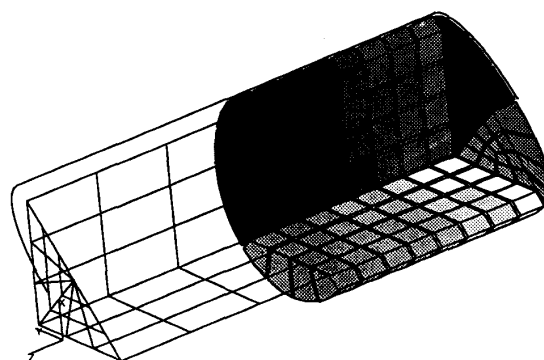


Figure 4: Part of mesh for PEC box example showing a fourth of the box, the generator of the BOR and a fourth of the subsequent mesh

The sampling rates on this BOR are $5s/\lambda$ along the generator and modes $-3, \dots, +3$, averaging a low $23s/\lambda^2$ over the $3\lambda^2$ surface area. The number of degrees of freedom of $\vec{J}$ (and $\vec{M}$) on Γ is $140 \approx 23s/\lambda^2 \cdot 3\lambda^2 \cdot 2$. There are 352 quadratic hexahedral finite elements providing a sampling rate of $10.5 \rightarrow 12.5s/\lambda$.

Computation of the monostatic cross sections involved first factoring a rank 12690 sparse matrix and back-substituting to generate 140 sparse solutions. Monostatic calculations involved a rank 140 dense matrix and an rank 140 block diagonal matrix (20 by 20 blocks). Figure 5 shows the monostatic cross section

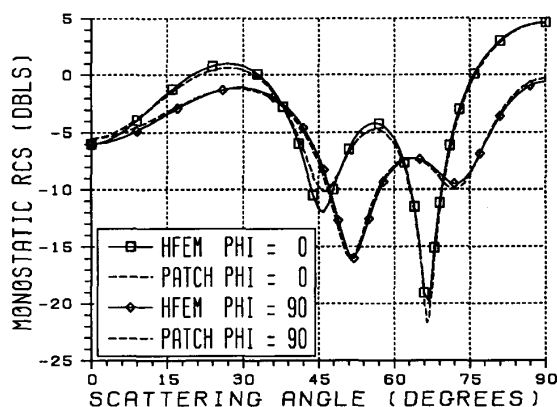


Figure 5: Monostatic scattering (dB/λ^2) from conducting box, $2m$ by $1/2m$ by $1/2m$ at 200 MHz.

comparison between this hybrid finite element method code and a three-dimensional MOM code PATCH from Sandia National Laboratory.

CONCLUSIONS

We have presented a new hybrid finite element method employing a unique BOR implementation utilizing Hermite cubic basis functions and the combined field integral equations. This formulation provides for efficient monostatic scattering calculations by combining the flexibility of a body of revolution mesh boundary, keeping the mesh boundary close to the scatterer, with a highly efficient BOR implementation, reducing the number of sparse solutions and the rank of the matrix used in scattering solutions. In addition, this method is not plagued by artificial resonances of the mesh boundary. This implementation has proven accurate and efficient on a large class of scatterers involving dielectric and magnetic materials, as well as the results shown here.

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